

Prediction

Diagnostics summary

Before fitting a model:

- + Check empirical logit plots; make transformations as needed
- + Check for multicollinearity with scatterplots / correlations between quantitative explanatory variables; if needed, choose a subset of variables or combine variables

After fitting a model:

- + Check quantile residual plots
- + Check for multicollinearity with VIFs
- + Check for influential points with Cook's distance
 - + If influential points are clearly an error, can remove
 - + Otherwise, report results with and without influential points

Using logistic regression

Class so far:

- + Model fitting
- + Model interpretation
- + Calculating predictions
- + Assessing model assumptions

What's next:

- + Assessing predictions
- + Variable selection
- + Inference

Warmup

Work on the warmup activity (handout), then we will discuss as a group

Class activity

Fitted model:

$$\log\left(\frac{\hat{\pi}_i}{1 - \hat{\pi}_i}\right) = 3.78 - 0.037Age_i - 2.52Male_i - 1.31Class2_i - 2.58Class3_i$$

What is the predicted probability of survival for a male, first-class passenger aged 20? What about for a male, second-class passenger aged 30?

① $\hat{\pi}_i = 0.63$

$$\frac{\exp(3.78 - 20(0.037) - 2.52)}{1 + \exp(3.78 - 20(0.037) - 2.52)}$$

② $\hat{\pi}_i = 0.24$

Making predictions with the Titanic data

- + For each passenger, we calculate $\hat{\pi}_i$ (estimated probability of survival)
- + But, we want to predict *which* passengers actually survive

How do we turn $\hat{\pi}_i$ into a binary prediction of survival / no survival?

$$\hat{\pi}_i = \text{predicted probability}$$
$$\hat{y}_i = \begin{cases} 1 & \hat{\pi}_i \geq 0.5 \\ 0 & \hat{\pi}_i < 0.5 \end{cases} \quad \leftarrow \text{threshold}$$

$\uparrow$
predicted outcome

Confusion matrix

```
m1 <- glm(Survived ~ Age + Sex + as.factor(Pclass),  
           data = titanic, family = binomial)
```

```
table(Prediction = m1$fitted.values > 0.5,  
      Truth = titanic$Survived)
```

		Truth	
		0	1
Prediction			
$\hat{y} = 0$ FALSE		356	83
$\hat{y} = 1$ TRUE		68	207

Correct predictions!

356: # passengers for which $y=0, \hat{y}=0$

83: # for which $y=1, \hat{y}=0$

68: $\hat{y}=1, y=0$

207: $\hat{y}=1, y=1$

Did we do a good job predicting survival?

Ideally, want accuracy, sensitivity, specificity, PPV all high

Confusion matrix

		Truth	
## Prediction		0	1
## FALSE		356	83
## TRUE		68	207

$\hat{Y}=0$

$\hat{Y}=1$

$Y=0$	$Y=1$
True negative (TN)	False negative (FN)
False positive (FP)	True positive (TP)

$$\text{Accuracy} : \frac{\# \text{ correct predictions}}{\# \text{ observations}} = \frac{TP + TN}{n} = \frac{356 + 207}{714} \approx 0.79$$

$P(\hat{Y}=Y)$

$$\text{Sensitivity} : P(\hat{Y}=1 | Y=1) = \frac{TP}{TP + FN} = \frac{207}{207 + 83} = 0.71$$

$$\text{Specificity} : P(\hat{Y}=0 | Y=0) = \frac{TN}{TN + FP} = \frac{356}{356 + 68} = 0.84$$

$$\text{Positive predictive value (PPV)} : P(Y=1 | \hat{Y}=1) = \frac{TP}{TP + FP} = \frac{207}{207 + 68} = 0.75$$

Class activity

Predictions with the SBA data:

https://sta214-f25.github.io/class_activities/ca_10.html

Class activity

```
m1 <- glm(Default ~ log(DisbursementGross) + Term +  
           as.factor(UrbanRural),  
           family = binomial, data = sba)  
  
table(Prediction = m1$fitted.values > 0.5,  
       Truth = sba$Default)
```

```
##           Truth  
## Prediction FALSE TRUE  
##      FALSE 3989  734  
##      TRUE  100  168
```

- + Accuracy =
- + Sensitivity =
- + Specificity =
- + PPV =

Class activity

```
##                Truth
## Prediction FALSE TRUE
##      FALSE  3989  734
##      TRUE   100  168
```

Is an accuracy of around 80% good?

Class activity

Changing thresholds:

```
table(Prediction = m1$fitted.values > 0.3,  
      Truth = sba$Default)
```

```
##           Truth  
## Prediction FALSE TRUE  
##      FALSE  3524  351  
##      TRUE   565  551
```

```
table(Prediction = m1$fitted.values > 0.7,  
      Truth = sba$Default)
```

```
##           Truth  
## Prediction FALSE TRUE  
##      FALSE  4089  902
```